REAL-WORLD BUSINESS ANALYTICS

Descriptive → Diagnostic → Predictive → Prescriptive Analysis

Objective: Apply the four types of analytics to five different real-world business situations and use the findings to support business decisions.

Scenario 1 – Online Grocery Delivery: Fresh Produce Waste

Business Problem: An online grocery service notices that some fruits and vegetables are frequently left unsold or discarded, while popular items sometimes run out before the end of the day.

Type of Analytics	Analysis / Answer
Descriptive Analytics – What happened?	Some fresh products had high wastage, while other products experienced stockouts. Sales and waste levels varied by day and by product.
Diagnostic Analytics – Why did it happen?	Demand may have been estimated inaccurately. Fresh produce has a short shelf life, and weather, weekends, promotions, and local preferences can affect orders.
Predictive Analytics – What might happen?	If ordering patterns continue, products with unstable demand may keep generating waste or stockouts. Forecasts using past orders, day of week, and season may improve estimates.
Prescriptive Analytics – What should we do?	Order smaller quantities more frequently for perishable items. Use demand forecasts, track expiry dates, discount items approaching expiry, and maintain safety stock for consistently popular products.

Scenario 2 – Coffee Shop: Changes in Customer Footfall

Business Problem: A coffee shop experiences heavy customer traffic in the morning but lower sales during the afternoon, leaving staff and equipment underused during quieter hours.

Type of Analytics	Analysis / Answer
Descriptive Analytics – What happened?	Sales and customer visits were highest during morning hours and lower during the afternoon. The number of drinks sold and average order value differed by time of day.
Diagnostic Analytics – Why did it happen?	Many customers buy coffee before work or school. Afternoon demand may be affected by nearby competitors, limited snack options, pricing, or customer routines.

Predictive Analytics – What might happen?	The morning rush is likely to continue on working days, while afternoon sales may remain lower unless customer behaviour changes. Weather and holidays may also affect footfall.
Prescriptive Analytics – What should we do?	Schedule more staff during peak hours and fewer during quiet periods. Introduce afternoon combo offers, promote snacks, and compare sales before and after promotions to measure their impact.

Scenario 3 – Clothing Retailer: Seasonal Inventory

Business Problem: A clothing retailer finds that winter jackets remain unsold when the season ends, while popular sizes of everyday clothing sell out quickly.

Type of Analytics	Analysis / Answer
Descriptive Analytics – What happened?	Some seasonal items remained in stock after demand declined, while certain sizes and styles had frequent stockouts. Inventory turnover varied across product categories.
Diagnostic Analytics – Why did it happen?	Purchasing quantities may not have matched customer demand. Seasonal timing, size preferences, changing fashion trends, and supplier lead times can create an imbalance.
Predictive Analytics – What might happen?	If purchasing patterns remain unchanged, the retailer may carry excess seasonal stock and miss sales of popular sizes. Historical sales by month, style, and size can help forecast demand.
Prescriptive Analytics – What should we do?	Use smaller initial orders and replenish fast-selling sizes. Start seasonal promotions before demand drops, review size-level sales, and transfer slow-moving items between stores where demand differs.

Scenario 4 – Video Streaming Service: Subscriber Cancellations

Business Problem: A streaming service sees an increase in customers cancelling their subscriptions after a free trial or during the first few months of paid membership.

Type of Analytics	Analysis / Answer
Descriptive Analytics – What happened?	Subscription cancellations increased among some customer groups. The cancellation rate and average subscription duration differed by plan and length of membership.

Diagnostic Analytics – Why did it happen?	Customers may not find enough content they enjoy, may consider the subscription too expensive, or may experience technical issues. Cancellation survey responses and viewing patterns can help investigate the causes.
Predictive Analytics – What might happen?	Customers with low viewing activity, repeated payment failures, or poor service experiences may be more likely to cancel. A model using historical subscriber behaviour could identify customers at risk.
Prescriptive Analytics – What should we do?	Recommend relevant content, improve playback reliability, remind trial users of useful features, and test retention offers for at-risk customers. Measure whether these actions reduce cancellations without giving unnecessary discounts.

Scenario 5 – Solar Energy Company: Installation Delays

Business Problem: A solar panel company receives more customer orders but struggles to complete installations on the promised dates because of equipment availability and scheduling problems.

Type of Analytics	**Analysis / Answer**
Descriptive Analytics – What happened?	The number of orders increased, but some installations were delayed. Completion time varied by location, installation team, and equipment availability.
Diagnostic Analytics – Why did it happen?	Delays may result from supplier shortages, weather, permit approvals, travel time, or overbooking of installation teams. Comparing delay reasons across projects can reveal recurring bottlenecks.
Predictive Analytics – What might happen?	If order volume grows without changes to supply and scheduling, average waiting times may increase. Past project durations, weather, and supplier lead times can help predict likely delays.
Prescriptive Analytics – What should we do?	Keep appropriate stock of frequently used components, confirm permits before scheduling, group installations by location, and use realistic time estimates. Inform customers early when a delay is likely.

Overall Conclusion

The four analytics stages provide a structured way to make business decisions. Descriptive analytics summarises what happened, diagnostic analytics explores why it happened, predictive analytics estimates what may happen next, and prescriptive analytics recommends actions. By applying these stages to grocery inventory, coffee shop footfall, clothing stock, streaming subscriptions, and solar installation scheduling, businesses can identify problems, plan ahead, and improve operational performance. The examples are illustrative; real decisions should be validated using the organisation’s actual data.